

# Global Facilities - Design & Construction Standard - CSA

## - Precast Concrete Works Specifications

|                |                   |                           |                |
|----------------|-------------------|---------------------------|----------------|
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[ECN History](#)

[Submit a Change Request](#)

### 1.0 Purpose

- 1.1 This document provides a Required Standard and Standard Guideline on Material and Design Specification that shall be followed and complied in the Design, Engineering and Construction for Precast Concrete Works Specifications as part of any green and/or brown field construction projects contract requirement.
- 1.2 The standard is applicable for any green and/or brown field construction projects as per required in the contract.
- 1.3 The document is intended to serve as a reference and guideline in addition to any Construction Drawings and Specification provided in the contract requirement including any required by local authority regulation and standard requirement as well as code of practice.
- 1.4 The established Specifications Standard for each locale shall also comply with Micron's in-house:
  - 1.4.1 RBA Code of Conduct requirements for Semiconductor Manufacturing
  - 1.4.2 Environmental, Health & Safety Standards
  - 1.4.3 Operational Requirements /SOP (Standard Operation Procedures)
  - 1.4.4 Security Protocol
- 1.5 The Specifications Standard to be adopted in each local shall not violate any local codes or by-laws and reflects the current semi-conductor industry's best practices.
- 1.6 Where there is the option to adopt either the MasterSpec or in-region

specifications standard, the AE (Architectural & Engineering Consultants) shall propose the most suitable solution based on safety, quality, time & cost to Micron's Designated Project Team (DPT) for review and acceptance. Otherwise, the more stringent option shall apply.

## 2.0 Scope

- 2.1 This document for precast concrete works specification covers all necessary related design, engineering, procurement, construction, installation and completion of piling, substructure, superstructure, and roof structure to full compliance of the contract requirement and required for the project. It extended to cover any green and/or brown field construction projects.
- 2.2 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.

## 3.0 Definitions and Acronyms

- 3.1 Definitions of acronyms used in this document are listed as follows:

| Term/Acronym                              | Definition / Description                                                                                                                                                                                                 |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FCT                                       | Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs                                                                                                 |
| GFTT                                      | Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards                                                                               |
| GFC                                       | Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities |
| Global Facilities Central Team            | This is used interchangeably with Innovation & Integration Team                                                                                                                                                          |
| Global Facilities Construction Team       | This is used interchangeably with Global Construction Team                                                                                                                                                               |
| Global Facilities Operations Support Team | This is used interchangeably with Global Facilities Operations Team                                                                                                                                                      |
| Global Facilities Planning Team           | This is used interchangeably with Global Facilities Line Planning Team                                                                                                                                                   |
| AE                                        | External Architectural & Engineering Consultants                                                                                                                                                                         |
| DPT                                       | Micron's Designated Project Team                                                                                                                                                                                         |
| GFCT                                      | Micron's Global Facilities Construction Team                                                                                                                                                                             |
| EHS                                       | Micron's Environmental Health & Safety Team                                                                                                                                                                              |
| QP/PE                                     | External Qualified Person (professional architect/engineer)                                                                                                                                                              |
| SME                                       | Micron's Subject Matter Expert                                                                                                                                                                                           |
| TM                                        | Micron's Team Members                                                                                                                                                                                                    |

## **4.0 Precast Concrete Structure**

- 4.1 To comprise architectural and structural precast concrete units conforming to the requirements of the tender drawings.

## **5.0 Scope of Works**

- 5.1 The design, manufacture and erection of the precast concrete units referred to above including all site works necessary to produce a fully stable structure able to withstand all temporary and permanent, vertical and lateral loading at all stages of erection and on completion.

## **6.0 Information to be Submitted With Tender**

- 6.1 General arrangement plans and sections at each floor level showing the layout and sizes of all precast concrete components together with the extent of all in-situ concrete provided.
- 6.2 A description of the construction including materials, method of erection and finishes. Any proposed departure from the tender documents must be clearly indicated.
- 6.3 A bar chart programme showing design, design approval, materials procurement, mould manufacture, production, erection and handover.
- 6.4 A full description of builder's work, special provisions and special attendances to be provided by others.
- 6.5 Limitations on cutting of holes in floor slabs, forming of chases and recesses and fixings to precast concrete components.

## **7.0 Contract Sum Analysis**

- 7.1 A categorized breakdown of the precast concrete units and ancillary work shall be given.

## **8.0 Programme Constraints**

- 8.1 Complete and hand over in an agreed sequence.

## **9.0 Design and Coordination**

- 9.1 Design and detail the work and provide complete production information (e.g. fabrication and installation drawings, design calculations, and specifications) based on the drawings, this Specification and other information provided, liaising with Consultant, Builder and others as necessary to help ensure co-ordination of the work with related building elements and services and to integrate the design of the structure with the total building. Provide appropriate representation at regular design/progress meetings as required by the Consultant.

- 9.2 Request additional information as necessary from the Consultant and/or Builder and provide information as necessary in time to meet the programme. Do not scale dimensions from drawings. Obtain from the Consultant any dimensions required but not given in figures on the drawings nor calculable therefrom. Report any discrepancies for clarification.
- 9.3 Submit or re-submit as appropriate 5 copies of the general arrangement drawings, 2 copies of calculations, 5 copies of unit details to the Consultant.
- 9.4 The Consultant will inspect the design/production information, record comments and returns to the Precaster. The Consultants will require fourteen days for such examination of design/production information.
- 9.5 Make any necessary amendments in accordance with any comments of the Consultant and without delay. Unless and until it is confirmed that re-submission is not required, re-submit for further checking and comment, and incorporate any necessary further amendments all as before.

## **10.0 Basis of Design**

- 10.1 Design the precast concrete structure, its components and connections in accordance with all relevant current Local or British Standards and to satisfy the requirements of the relevant Building Regulations.
- 10.2 Design the precast concrete structure as a stable structure able to withstand all temporary and permanent, vertical and lateral loading without additional framing or stiffening from cladding, block walls, partitions, roof bracing, etc. Accept responsibility for the stability and structural integrity of the precast concrete structure during erection and support as necessary.
- 10.3 Design precast concrete structure to withstand the effects of accidental damage and to satisfy relevant Building Regulations.
- 10.4 Design beam/column junctions as rigid joint frame.

## **11.0 Loading Criteria**

- 11.1 Design the structure for dead and imposed loads as stipulated on plan.
- 11.2 Wind loading and notional horizontal load.

The fundamental value of the basic wind velocity ( $V_{b,0}$ ) is 20 m/s, in accordance with SNA to Eurocode 1; NA to SS EN 1994-1-4 or equivalent local code.

Notional horizontal load in accordance with SS EN 1992 or equivalent requirement shall be considered.
- 11.3 Seismic Loading as specified in the local or equivalent international requirements.
- 11.4 Miscellaneous loads like snow load as per project location and local code requirements.
- 11.5 Stability and robustness requirements in accordance with SS EN1992 or equivalent code requirements shall be considered.

## **12.0 Deflection**

- 12.1 Limit vertical deflections under imposed load to  $L/500$  or 15 mm whichever is the lesser, where  $L$  is the span of the beam or slab. Limit lateral deflection in any one storey under wind load to  $H/500$  where  $H$  is the floor-to-floor storey height.

## **13.0 Fire Resistance**

- 13.1 The structure is to be designed for a fire resistance of 2 hours.

## **14.0 Integrity**

- 14.1 The precast concrete components must be reinforcement as necessary to allow for the effects of service loads, handling stresses, creep, shrinkage and deflection.

## **15.0 Calculations**

- 15.1 Provide design calculations and details together with any relevant test data to enable design assumptions to be verified. Such calculations, details and data are also to be sufficient to satisfy the requirements of the Building Control Authority. Notwithstanding the above, the Precaster is to remain responsible for the design and construction of the structure of the building insofar as it is designed and constructed by them. Assistance in the checking of drawings and calculations will be given by the Consultant but any failure to detect inconsistencies or errors will not relieve the Precaster of responsibility.

## **16.0 Durability**

- 16.1 Subject to other specified requirements, the quality of concrete and nominal cover to reinforcement (including secondary reinforcement) must be related to the following environment : mild/moderate.14.2 Wind loading and notional horizontal load
- 16.2 Increase the nominal cover to allow for depth of exposure on exposed aggregate finishes.
- 16.3 If galvanized or stainless steel reinforcement is to be used submit proposals for grade of concrete and nominal cover for approval.

## **17.0 Concrete Mixes Generally**

- 17.1 Constituent materials, composition of mixes, production of concrete, information to be provided, sampling, testing and compliance: all to be in accordance with BS 8500, BS EN 206-1 and local Standards unless otherwise specified.
- 17.2 The mix constituents to comply with the limitations given in SS EN 1992 in

respect to the use and location of the elements.

17.3 Rate of sampling for compressive strength testing: not less than one samples for each day of use.

17.4 Concrete grade of all precast elements shall not be less than Grade C32/40.

## **18.0 Correlated Records**

18.1 Correlated records must be maintained for each mix type, including:

- Composition of the mix, including any admixtures.
- Results of slump and other tests carried out at the casting yard.
- Laboratory test reports, including cube identification numbers.

## **19.0 Test Laboratory**

19.1 All compressive strength testing to be carried out by Accredited laboratory. Submit the name of the selected laboratory to the Consultant as soon as possible and in any case before commencing manufacture.

## **20.0 Generally**

20.1 Provide evidence of satisfactory quality control procedures and their effective implementation. Allow access by the Consultant or nominated representative to all manufacturing facilities during working hours.

The supply sources/factory of precast elements shall be from Singapore.

## **21.0 Moulds**

21.1 Moulds must be so constructed that casting deviations can be controlled to give compliance with clause 21.1 Construct to prevent loss of grout. Design to permit demoulding without damage to the units. Coat evenly with a suitable release agent. Maintain in clean, sound condition and inspect carefully for defects before each re-use.

## **22.0 Plain Smooth Finish**

22.1 Produce an even finish with a sheet material.

22.2 Abrupt irregularities to be not greater than 5mm. Gradual irregularities, expressed as maximum permissible deviation from a 1m straight edge, to be no greater than 5mm.

22.3 Blowholes less than 1mm in diameter will be permitted but otherwise surface to be free from voids, honeycombing, segregation and other large defects. Precaster shall fill-up defects with approved material.

22.4 Making good: projecting fins are to be removed and rubbed down with a carborundum stone but otherwise the finish is to be left as struck.

### **23.0 Smooth Floated Finish**

- 23.1 Use a hand float, skip or power float to give an even surface with minimal ridge or steps.

### **24.0 Manufacturing Accuracy**

- 24.1 Finished dimensions of precast concrete components to comply with the manufacturing tolerances given in Tables 1 and 2 of BS 5606 or equivalent.
- 24.2 Check the overall dimensions, straightness, squareness, twist and flatness of the mould(s) immediately before each re-use, and of each component as soon as possible after demoulding. Make adjustments to moulds as necessary.
- 24.3 Precaster shall check and ensure that the individual and combined deviations established (in accordance with clause 9 of BS 5606 or equivalent) meet the needs of the design and construction.

### **25.0 Camber**

- 25.1 Reinforced components are to be cambered and agreed in advance with the Consultant. For prestressed concrete components, the predicted camber in accordance with Clause 6.2.7.4 of SS CP 65:Part 1 or equivalent shall not be greater than 25mm unless otherwise agreed.
- 25.2 Variation in camber between adjacent floor units of the same span and loading shall, except where otherwise qualified, not exceed half the predicted value.

### **26.0 Fixings**

- 26.1 Lifting and adjustment devices, which are away from the external face and edges and recessed for subsequent filling with mortar may be of galvanized steel. Other devices must be of a suitable type of stainless steel or non-ferrous metal. Prevent bimetallic corrosion and staining.

### **27.0 Reinforcement**

- 27.1 Unless otherwise specified or equivalent code requirements:
- Reinforcement to SS 560 and SS 561, cut and bent to BS 8666
  - Prestressing strand to BS 5896
- 27.2 Galvanized reinforcement: galvanized to BS EN ISO 1461 or equivalent code requirements after cutting, chromate treated.
- 27.3 Stainless steel reinforcement: to BS 6744, type 304 S31 or alternative to Consultant's approval.
- 27.4 Surface condition: to Clause 7.4 of SS CP 65 or equivalent code requirements at time of placing.

- 27.5 Fix accurately and securely using tying wire, approved steel clips, or tack welding to BS EN ISO 17660 or equivalent code requirements if permitted. Wire or clips must not encroach into the concrete cover greater than their thickness or diameter.

## **28.0 Casting and Curing**

- 28.1 Thoroughly compact concrete by mechanical vibration.
- 28.2 Do not demould components prematurely.
- 28.3 Prevent damage to and distortion of immature components from movement, vibration, overloading, physical shock, rapid cooling and thermal shock.
- 28.4 Do not deliver components to site without prior approval, until at least three days after casting.

## **29.0 Records**

- 29.1 Keep complete records for each component including the following information:
- An identification number
  - Correlation with record of mixes
  - Date of manufacture
  - Dates and results of all tests, checks and inspections
  - Dimensions related to specified levels of accuracy
- 29.2 Records to be available for inspection on request.

## **30.0 Identification**

Mark all components in stock with contract reference.

## **31.0 Erection, Fixing, Jointing**

- 31.1 To be carried out by the Precaster who must:
- Provide clear and comprehensive instructions and ensure that they are understood by the site operatives.
  - Provide adequate site supervision by suitably skilled person(s).

## **32.0 Protection**

- 32.1 Prevent overstressing of components during transit, handling, storage and fixing. Lift components at designed lifting points only, using special lifting devices and cradles as necessary.
- 32.2 Store components on level bearers clear of the ground and separate with spacers at designed locations.
- 32.3 Prevent damage to components.

### **33.0 Prestressed Floor Slabs**

33.1 Inspect on arrival on site and prior to grouting or in-situ concreting. Do not erect slabs which exhibit any of the following:

- Excessive transport or handling damage
- Manufacturing defects
- Excessive cracks
- Excessive areas of lean or honeycombed concrete
- Excessive camber
- Tendon 'pull-in' exceeding 5mm

### **34.0 Holding-Down Bolts**

34.1 Supply holding-down bolts to the Builder to suit programme requirements. Make allowance for holding down bolts positioned within an accuracy of  $\pm 6\text{mm}$  by the Builder.

### **35.0 Accuracy of Erection**

35.1 Construct the works to levels of accuracy as given in Table 1 of BS 5606 or local code requirements.

35.2 Do not rectify work, which fails to meet the specified levels of accuracy without approval.

35.3 Alignment of precast units/components are of critical importance, both transversely and longitudinally

### **36.0 Foundation**

36.1 Before starting erection, inspect the foundations including holding down bolts, column starter bars and dowel pockets for line and level, and confirm accuracy in writing to the Consultant.

### **37.0 In-Situ Concrete**

37.1 Where in-situ concrete is designed into the structure, provide details of mixes and production. Make cubes for each day of concreting, test and submit to the Consultant.

### **38.0 Grout**

38.1 Provide details of the mix and production. Make cubes as required, test and submit written reports to the Consultant.

### **39.0 Damaged Units**

- 39.1 Do not repair without approval. Such approval will not be given where the components are badly damaged or where the proposed repair would impair performance. Immediately remove rejected units from the site.

### **40.0 Site Welding**

- 40.1 All site fillets welding must be in accordance with BS EN ISO 17660 or equivalent code requirements as appropriate. Take due account of the embedment of steel fittings in concrete where this will affect welding. Site welds must be inspected, visually 100% and MPI 10% as a minimum requirement. All welds failing visual inspection must be MPI tested. All welding defects must be rectified prior to work being covered up, and inspection records must be available as required. Procedures for butt welding to be submitted to the Consultant for approval.

### **41.0 Bearing Strips/Pads**

- Neoprene bearing strips/pads shall be provided.

### **42.0 Supervision of Precast Structural Elements**

- 42.1 Construction of precast structural elements manufactured off-site in Singapore or overseas, shall be supervised by accredited site supervisors under the direction of a separate QP (Supervision) appointed by the licensed builder/specialist builder.
- 42.2 A separate permit to commence works is also required for the precast structural elements.
- 42.3 The Builder's appointed QP (Supervision) shall submit a certificate of supervision of the precast structural elements on completion of the project.

### 43.0 Document Control

43.1 Approval: GLOBAL\_FAC\_GFTT\_APPR

43.2 Notification:

GLOBAL\_FAC\_MANAGERS  
GLOBAL\_FAC\_COMMUNICATIONS  
GLOBAL\_FAC\_CONSTRUCTION  
GLOBAL\_FAC\_NOTIFY  
FCG\_F4\_FAC\_NOTIFY  
FCG\_F6\_FAC\_NOTIFY  
FCG\_F10\_FAC\_NOTIFY  
FCG\_F11\_FAC\_NOTIFY  
FCG\_F15\_FAC\_NOTIFY  
FCG\_F16\_FAC\_NOTIFY  
FCG\_MMP\_FAC\_NOTIFY  
FCG\_MMY\_FAC\_NOTIFY  
FCG\_MSB\_FAC\_NOTIFY  
FCG\_MXA\_FAC\_NOTIFY  
FCG\_MTB\_FAC\_NOTIFY

43.3 Notification:

All Micron Team Members at all Sites

43.4 Retention

Adherence to the requirements specified in

[Global Facilities - Document Control Procedures](#)

43.5 Review

Biennially (two years)

### 44.0 Revision History

| Revision # | Section Changed | Details of Changes | Date       |
|------------|-----------------|--------------------|------------|
| 0          | NA              | New Document       | 07/06/2022 |